

Experiment No. - 1

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Aim/Overview of the practical: To develop a NodeJS program that calculates the salary of an employee using basic arithmetic operations and user-defined functions.

Objective:

After performing this experiment, students will be able to:

- Understand basic NodeJS program structure
- Use variables and operators in NodeJS
- Perform arithmetic calculations
- Implement user-defined functions
- Execute NodeJS programs using terminal

1. Code for experiment:

```
const readline = require("readline");

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

rl.question("Enter Employee Name: ", function(name) {
  rl.question("Enter Employee ID: ", function(id) {
    rl.question("Enter Department: ", function(dept) {
      rl.question("Enter Basic Salary: ", function(salary) {
        salary = parseFloat(salary);
        let hra = salary * 0.20;
        let da = salary * 0.10;
        let gross = salary + hra + da;
```

```

console.log("\n----- EMPLOYEE SALARY SLIP -----\n");
console.log("Name      : " + name);
console.log("ID        : " + id);
console.log("Department : " + dept);
console.log("Basic Pay   : " + salary.toFixed(2));
console.log("HRA (20%)   : " + hra.toFixed(2));
console.log("DA (10%)    : " + da.toFixed(2));
console.log("Gross Pay   : " + gross.toFixed(2));

    rl.close();
  });
});
});
});

```

Output:

```

PS C:\Users\salon\OneDrive\Desktop\Backend\SignupProject\Salaryproject> node .\salary.js
Name      : Saloni
ID        : 123
Department : HR
Basic Pay   : 40000.00
HRA (20%)   : 8000.00
DA (10%)    : 4000.00
Gross Pay   : 52000.00
PS C:\Users\salon\OneDrive\Desktop\Backend\SignupProject\Salaryproject>

```

2.Code for experiment:

```

const http = require("http");
const querystring = require("querystring");

function grossSalary(basic, hra, da) {
  return basic + hra + da;
}

function netSalary(gross, pf, tax) {
  return gross - (pf + tax);
}

const server = http.createServer((req, res) => {
  if (req.method === "GET") {

    res.writeHead(200, { "Content-Type":
"text/html; charset=UTF-8" });
    res.end(`

```

```

<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Employee Salary Calculator</title>

<style>

body{
font-family:Arial;
background:linear-gradi-
ent(135deg,#4facfe,#00f2fe);
display:flex;
justify-content:center;
align-items:center;
height:100vh;
margin:0;
}

```

```
.box{
background:white;
padding:30px;
border-radius:15px;
width:380px;
box-shadow:0 0 20px gray;
}
h2{
text-align:center;
color:#0066cc;
}
input{
width:100%;
padding:12px;
margin:10px 0;
box-sizing:border-box;
}
button{
width:100%;
padding:12px;
background:#0066cc;
color:white;
font-size:18px;
border-radius:8px;
cursor:pointer;
}
button:hover{
background:#004999;
}
</style>
</head>
<body>
<div class="box">
<h2>Employee Salary Form</h2>
<form method="POST">
<input type="number" name="basic" place-
holder="Basic Salary" required>
<input type="number" name="hra" place-
holder="HRA" required>
<input type="number" name="da" place-
holder="DA" required>
<input type="number" name="pf" place-
holder="PF" required>
<input type="number" name="tax" place-
holder="Tax" required>
<button>Calculate Salary</button>
</form>
```

```
</div>
</body>
</html>
`);
}
else if(req.method==="POST"){
let body="";
req.on("data",chunk=>{
body+=chunk.toString();
});

req.on("end",()=>{

const
data=querystring.parse(body);

let basic=parseFloat(data.basic);
let hra=parseFloat(data.hra);
let da=parseFloat(data.da);
let pf=parseFloat(data.pf);
let tax=parseFloat(data.tax);

let gross=grossSal-
ary(basic,hra,da);
let net=netSalary(gross,pf,tax);

res.writeHead(200,{"Content-
Type":"text/html; charset=UTF-8"});

res.end(`
<!DOCTYPE html>
<html>

<head>
<meta charset="UTF-8">
<style>

body{
font-family:Arial;
background:#f2f2f2;
display:flex;
justify-content:center;
align-items:center;
height:100vh;
}
```

```
.card{
background:white;
padding:30px;
border-radius:15px;
width:420px;
box-shadow:0 0 15px gray;
}
h2{
color:green;
text-align:center;
}
p{
font-size:20px;
}
a{
display:block;
text-align:center;
background:#0066cc;
color:white;
padding:10px;
border-radius:8px;
text-decoration:none;
}
```

```
</style>
</head>
<body>
<div class="card">
<h2>Salary Details</h2>
<p><b>Basic Salary :</b> ₹${basic}</p>
<p><b>HRA :</b> ₹${hra}</p>
<p><b>DA :</b> ₹${da}</p>
<p><b>Gross Salary :</b> ₹${gross}</p>
<p><b>PF :</b> ₹${pf}</p>
<p><b>Tax :</b> ₹${tax}</p>
<p><b>Net Salary :</b> ₹${net}</p>
<br>
<a href="/">Calculate Again</a>
</div>
</body>
</html>
`); });
    }
});
server.listen(5000,()=>{
console.log("Server Running at
http://localhost:5000");
});
```

OUTPUT:

Employee Salary Form

Calculate Salary

Salary Details

Basic Salary : ₹40000

HRA : ₹8000

DA : ₹4000

Gross Salary : ₹52000

PF : ₹1200

Tax : ₹2000

Net Salary : ₹48800

Calculate Again

Learning Outcomes (What I Have Learnt)

- Understand the basics of **Node.js** and server-side JavaScript programming.
- Create a simple HTTP server using the built-in `http` module.
- Design an HTML form to collect user input through a web browser.
- Process user input on the server and perform salary calculations.
- Calculate **HRA (20%)**, **DA (10%)**, and **Gross Salary** using JavaScript.
- Generate and display dynamic HTML content in the browser.